

KinSync — Requirements & Acceptance Criteria (spec.md)

Project: KinSync — Passive Daily Pattern Monitor for Elderly Living Alone **Version:** 0.1 (Phase-1 draft) **Last updated:** 12 Sep 2026 **Status:** Draft for Review 2

1. Problem Statement

Elderly individuals living alone are at risk of medical emergencies or falls going unnoticed for hours. Existing safety apps rely either on **active check-ins** (which lapse when the person forgets, is unwell, or resists the extra chore) or on **continuous location tracking** (which many elderly users experience as invasive and refuse to enable, undermining the very safety net it's meant to provide).

2. Proposed Solution (Summary)

KinSync passively learns an individual's normal daily digital-activity signature — phone-unlock times, typical app-usage windows, and coarse movement state — over an initial baseline period, then flags deviations from that pattern (e.g., no activity by the person's usual check-in time) to designated family contacts. Only derived signals ever leave the device; raw activity logs, content, and continuous location are never transmitted.

3. Goals

- **G1:** Reduce time-to-notice for a possible medical emergency or fall, without requiring the elder to actively “check in.”
- **G2:** Preserve the elder's dignity and privacy — no raw location, content, or continuous surveillance leaves the device.
- **G3:** Give family/caregivers timely, low-noise alerts they can trust (low false-positive rate).
- **G4:** Remain robust even if the elder's phone is offline, powered off, or unresponsive — the scenario the app most needs to catch.
- **G5:** Ship as a native Android solution with a self-hosted backend, giving the student team full ownership of both the mobile and server layers for learning purposes.

4. Non-Goals (v1)

- No medical diagnosis or clinical claims of any kind.
- No fall detection via accelerometer/gyroscope (candidate for future work, not v1).
- No wearable device integration.
- No iOS support (Android-only by design; see NFR-5).
- No continuous raw GPS tracking at any point.
- No multi-language UI in v1 (English only).

5. Stakeholders / User Roles

Role	Description
Elder	The person being passively monitored. Primary user of the “Elder” mode of the app.
Caregiver / Family member	One or more people who receive alerts and monitor status. Primary user of the “Caregiver” mode of the app.
Project guide / review panel	Evaluates the project against this spec and the review timeline.

6. Representative User Stories

- As an **elder**, I want the app to quietly notice if something’s wrong without me having to remember to check in, so that help can be found even if I can’t ask for it myself.
- As an **elder**, I want to see exactly what the app is tracking about me, so I can trust it isn’t spying on me.
- As an **elder**, I want to pause monitoring when I’m traveling, so I don’t get flagged for a change in routine I already know about.
- As a **caregiver**, I want to be notified quickly if my parent’s normal routine is disrupted, so I can check on them without having to call every day.
- As a **caregiver**, I want alerts to be rare and meaningful, so I don’t start ignoring them.

7. Functional Requirements

FR-1 — Onboarding & Pairing

- **FR-1.1** The app shall support two roles selectable at first launch: *Elder* and *Caregiver*.
- **FR-1.2** An Elder profile shall be able to generate a unique, time-limited pairing code.
- **FR-1.3** A Caregiver shall be able to redeem a pairing code to link their app to an elder’s profile.
- **FR-1.4** The system shall support one elder linked to up to three caregivers in v1.
- **FR-1.5** An elder shall be able to view and revoke linked caregivers at any time.

FR-2 — Passive Data Collection (on-device)

- **FR-2.1** The app shall capture screen-unlock events via a `BroadcastReceiver` (`ACTION_USER_PRESENT`, screen on/off).
- **FR-2.2** The app shall query app-usage windows via `UsageStatsManager`, requesting the special `PACKAGE_USAGE_STATS` permission with a clear, plain-language rationale screen before requesting it.
- **FR-2.3** The app shall capture coarse movement state (still / walking / in-vehicle) via the Activity Recognition API, without ever accessing raw GPS coordinates.
- **FR-2.4** All raw collected events shall be stored only in a local on-device database; raw events must never be transmitted off-device.
- **FR-2.5** The elder shall be able to view a plain-language log of what has been collected (transparency screen).

- **FR-2.6** The elder shall be able to pause/snooze collection for a defined period (e.g., a “traveling” mode).

FR-3 — Baseline Learning

- **FR-3.1** The app shall compute a rolling baseline (default 14-day window, configurable) of typical first-unlock time, active windows, and unlock frequency.
- **FR-3.2** The baseline shall use descriptive statistics (mean/median, standard deviation) in v1. A baseline is not considered “ready” until a minimum of 7 days of data exist.
- **FR-3.3** The app shall recompute the baseline periodically (e.g., nightly) as new days roll into the window.

FR-4 — Deviation Detection & Escalation

- **FR-4.1** The app shall run a periodic local check comparing the current day’s activity against the baseline.
- **FR-4.2** If no activity is detected within the elder’s expected check-in window, the app shall show an in-app nudge (“Everything OK? Tap to confirm”) before escalating.
- **FR-4.3** If the nudge is unacknowledged within a configurable grace period (default 30 minutes), the day’s heartbeat shall be marked “missed.”
- **FR-4.4** The backend scheduler shall independently track each elder’s expected check-in window (from a derived summary only) and shall trigger escalation if no heartbeat/acknowledgement is received within that window — **even if the device is offline, powered off, or unresponsive.**
- **FR-4.5** Escalation shall notify all linked caregivers via push notification (SMS fallback optional, later phase).
- **FR-4.6** An escalated alert shall remain visible to caregivers until explicitly marked resolved.

FR-5 — Notifications

- **FR-5.1** The caregiver app shall register a push-notification device token with the backend.
- **FR-5.2** The backend shall send a push notification containing minimal alert context (elder name, alert type, time) to all linked caregivers on escalation.
- **FR-5.3** (*Later phase*) The backend may send an SMS fallback via a configured provider for caregivers without the app installed.

FR-6 — Caregiver Dashboard

- **FR-6.1** The caregiver app shall display current status (normal / nudged / escalated) for each linked elder.
- **FR-6.2** The caregiver app shall display an aggregated activity trend without exposing raw time-stamps or app names.
- **FR-6.3** A caregiver shall be able to acknowledge/resolve an active alert.

FR-7 — Privacy & Consent Controls

- **FR-7.1** An elder must explicitly grant onboarding consent describing what is collected before the app begins collection.
- **FR-7.2** Only derived signals (a heartbeat and an expected check-in window) leave the elder’s device; raw activity logs never leave the device.

- **FR-7.3** An elder can revoke consent at any time, which stops collection and unpairs all caregivers.

8. Non-Functional Requirements

ID	Requirement
NFR-1 Privacy	No raw location, app-name-level detail, or content ever leaves the elder's device. Only derived heartbeat/window summaries are transmitted.
NFR-2 Reliability	The escalation mechanism must function as a "dead-man's switch" — it must trigger correctly even if the elder's phone is offline, powered off, or the app has been killed by OEM battery management, within the defined heartbeat cadence.
NFR-3 Battery	Background collection should not perceptibly reduce the elder's phone's battery life beyond an agreed threshold (target: <5% additional daily drain, to be measured in testing).
NFR-4 Security	All client-server traffic over TLS. Device-bound bearer tokens for authentication. VM hardened with a firewall, key-only SSH, and automated security updates.
NFR-5 Compatibility	Android only. Minimum SDK to be finalized by the team in Phase-1 based on UsageStatsManager/Activity Recognition availability; Google Play Services required.
NFR-6 Usability	Elder-facing UI must use large text, minimal steps per screen, and avoid technical jargon.
NFR-7 Maintainability	Code organized into the modules defined in <code>plan.md</code> , each independently testable.

9. Acceptance Criteria (key flows)

Pairing - *Given* an elder has generated a pairing code, *when* a caregiver enters that code within its validity window, *then* the caregiver's app is linked to the elder's profile and both apps reflect the new pairing.

Baseline readiness - *Given* fewer than 7 days of collected data, *when* the deviation detector runs, *then* it shall take no action (baseline not yet ready). - *Given* 7 or more days of collected data, *when* the deviation detector runs, *then* it shall compare today's activity against the computed baseline.

Nudge before escalation - *Given* the elder's expected check-in window has passed with no detected activity, *when* the deviation detector runs, *then* the elder shall receive an in-app nudge before any family notification is sent.

Dead-man's-switch escalation - *Given* no heartbeat has been received by the backend within an elder's expected window, *when* the scheduler runs its periodic check, *then* an alert shall be created and all linked caregivers notified — regardless of whether the elder's device is reachable at that moment.

Consent revocation - *Given* an elder revokes consent, *when* the revocation is confirmed, *then* all local collection shall stop immediately and all caregiver pairings shall be removed.

10. Assumptions & Constraints

- One elder may be linked to at most three caregivers in v1.
- The elder's device has Google Play Services (required for Activity Recognition and push notifications).
- The elder's device has internet connectivity at least once within each expected check-in window for a normal (non-escalated) day; the offline/unresponsive case is exactly what FR-4.4 is designed to catch.
- Backend infrastructure is a self-hosted Ubuntu 26.04 VM on Hetzner, chosen deliberately over a managed BaaS so the team gains backend engineering experience (see `plan.md`).

11. Out of Scope for v1

- Fall detection via accelerometer/gyroscope.
- Wearable device integration.
- Multi-language support.
- iOS support.
- More than three caregivers per elder.
- Audio or video monitoring of any kind.

12. Glossary

Term	Meaning
Baseline	A statistical model of an elder's typical daily activity pattern, computed on-device.
Heartbeat	A minimal, periodic signal sent from the elder's device to the backend, confirming activity is within the expected pattern.
Deviation	A detected departure from the elder's baseline pattern (e.g., no unlock by the usual time).
Escalation	The process of notifying caregivers after a deviation has gone unacknowledged past the grace period.
Pairing code	A short-lived code used to link a caregiver's app to an elder's profile.
Dead-man's switch	A safety pattern where the <i>absence</i> of an expected signal (not the presence of an explicit alarm) triggers a response.
Grace period	The window given to the elder to acknowledge a nudge before escalation proceeds.